Section 1.3 Complex numbers & roots of unity

复数与单位根

[AIME-2007-I-Q03] | Part A (Q1-Q5)

For $z = 9 + bi$ with $b > 0$, the imaginary parts of z^2, z^3 are equal. Find b.

$z = 9 + bi, b > 0$，若 z^2, z^3 虚部相等，求 b。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 1.3 Complex numbers & roots of unity

复数与单位根

[AIME-2009-I-Q02] | Part A (Q1-Q5)

A complex z has imaginary part 164 and satisfies $z/(z+n) = 4i$ for a positive integer n. Find n.

复数 z 的虚部为 164，且正整数 n 满足 $z/(z+n) = 4i$，求 n。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 1.3 Complex numbers & roots of unity

复数与单位根

[AIME-2018-II-Q05] | Part A (Q1-Q5)

Complex numbers satisfy $xy = -80 - 320i, yz = 60, zx = -96 + 24i$. If $x + y + z = a + bi$ with real a, b, find $a^2 + b^2$.

复数满足 $xy = -80 - 320i, yz = 60, zx = -96 + 24i$。若 $x + y + z = a + bi$，a, b 为实数，求 $a^2 + b^2$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 1.3 Complex numbers & roots of unity

复数与单位根

[AIME-2022-I-Q04] | Part A (Q1-Q5)

Set $w = (\sqrt{3} + i)/2$ and $z = (-1 + i\sqrt{3})/2$. Count integer pairs (r, s) with $1 \le r, s \le 100$ for which $iw^r = z^s$.

令 $w = (\sqrt{3} + i)/2, z = (-1 + i\sqrt{3})/2$。求满足 $iw^r = z^s$ 且 $1 \le r, s \le 100$ 的整数有序对数目。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 1.3 Complex numbers & roots of unity

复数与单位根

[AIME-2026-I-Q05] | Part A (Q1-Q5)

For $AB = 1$, rotate A counterclockwise through acute θ about B to $A\prime$, then rotate B clockwise through θ about $A\prime$ to $B\prime$. Given $AB\prime = 4/3$, write $\cos\theta = m/n$ in lowest terms and find $m + n$.

$AB = 1$，将 A 绕 B 逆时针转锐角 θ 得 $A\prime$，再将 B 绕 $A\prime$ 顺时针转 θ 得 $B\prime$。若 $AB\prime = 4/3$，将 $\cos\theta$ 化为最简分数 m/n，求 $m + n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 1.3 Complex numbers & roots of unity

复数与单位根

[AIME-2000-II-Q09] | Part B (Q6-Q10)

A complex number satisfies $z + z^{-1} = 2\cos 3^\circ$. Find the smallest integer strictly greater than $z^{2000} + z^{-2000}$.

复数满足 $z + z^{-1} = 2\cos 3^\circ$。求严格大于 $z^{2000} + z^{-2000}$ 的最小整数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 1.3 Complex numbers & roots of unity

复数与单位根

[AIME-2005-I-Q06] | Part B (Q6-Q10)

Let P be the product of the nonreal roots of $x^4 - 4x^3 + 6x^2 - 4x = 2005$. Find $\lfloor P \rfloor$.

令 P 为方程 $x^4 - 4x^3 + 6x^2 - 4x = 2005$ 的所有非实根之积，求 $\lfloor P \rfloor$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 1.3 Complex numbers & roots of unity

复数与单位根

[AIME-2005-II-Q09] | Part B (Q6-Q10)

Count $1 \leq n \leq 1000$ for which $(\sin t + i\cos t)^n = \sin(nt) + i\cos(nt)$ holds for every real t.

求 $1 \leq n \leq 1000$ 中使 $(\sin t + i\cos t)^n = \sin(nt) + i\cos(nt)$ 对所有实数 t 成立的整数个数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 1.3 Complex numbers & roots of unity

复数与单位根

[AIME-2008-II-Q09] | Part B (Q6-Q10)

Start at $(5,0)$ and repeat 150 times: rotate counterclockwise 45° about the origin, then translate ten units right. If the endpoint is (p,q), find $\lfloor |p|+|q| \rfloor$.

从 $(5,0)$ 出发，重复一百五十次“绕原点逆时针转四十五度，再向右平移十”，终点为 (p,q)，求 $\lfloor |p|+|q| \rfloor$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 1.3 Complex numbers & roots of unity

复数与单位根

[AIME-2010-II-Q07] | Part B (Q6-Q10)

A monic cubic with real coefficients has roots $w+3i, w+9i, 2w-4$ for some complex w. If it is z^3+az^2+bz+c, find $|a+b+c|$.

某首一实系数三次多项式的根为 $w+3i, w+9i, 2w-4$。若其为 z^3+az^2+bz+c，求 $|a+b+c|$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 1.3 Complex numbers & roots of unity

复数与单位根

[AIME-2011-II-Q08] | Part B (Q6-Q10)

Let $z_1, \ldots, z_{12}$ be all roots of $z^{12} = 2^{36}$. Independently choose $w_j = z_j$ or iz_j. The largest real part of $\sum w_j$ is $m + \sqrt{n}$. Find $m + n$.

令 $z_1, \ldots, z_{12}$ 为 $z^{12} = 2^{36}$ 的全部根。每项独立选择 $w_j = z_j$ 或 iz_j。若 $\sum w_j$ 实部的最大值为 $m + \sqrt{n}$，求 $m + n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 1.3 Complex numbers & roots of unity

复数与单位根

[AIME-2012-I-Q06] | Part B (Q6-Q10)

Complex numbers obey $z^{13} = w, w^{11} = z$, and $\operatorname{Im} z = \sin(m\pi/n)$ for coprime integers $0 < m < n$. Find n.

复数满足 $z^{13} = w, w^{11} = z$，且 $\operatorname{Im} z = \sin(m\pi/n)$，其中整数 $0 < m < n$ 互质。求 n。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 1.3 Complex numbers & roots of unity

复数与单位根

[AIME-2012-II-Q06] | Part B (Q6-Q10)

Among complex $z = a + bi$ with $|z| = 5, b > 0$, choose one maximizing the distance from $(1 + 2i)z^3$ to z^5. If $z^4 = c + di$, find $c + d$.

复数 $z = a + bi$ 满足 $|z| = 5, b > 0$，取使 $(1 + 2i)z^3$ 与 z^5 的距离最大的 z。若 $z^4 = c + di$，求 $c + d$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 1.3 Complex numbers & roots of unity

复数与单位根

[AIME-2012-II-Q08] | Part B (Q6-Q10)

Nonzero complex z, w satisfy $z + 20i/w = 5 + i$ and $w + 12i/z = -4 + 10i$. Find the least $|zw|^2$.

非零复数满足 $z + 20i/w = 5 + i, w + 12i/z = -4 + 10i$，求 $|zw|^2$ 的最小值。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 1.3 Complex numbers & roots of unity

复数与单位根

[AIME-2013-I-Q10] | Part B (Q6-Q10)

For nonzero integers a, b, r, s, the cubic $x^3 - ax^2 + bx - 65$ has root $r + si$. For each distinct possible coefficient pair (a, b), take the sum of its three roots. Find the sum of these values over all pairs.

非零整数 a, b, r, s 使 $x^3 - ax^2 + bx - 65$ 有根 $r + si$。对每个不同的可行系数对 (a, b)，取其三根之和，再将所有这些值相加，求总和。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 1.3 Complex numbers & roots of unity

复数与单位根

[AIME-2014-I-Q07] | Part B (Q6-Q10)

Complex w, z satisfy $|w| = 1, |z| = 10$. Let $\theta = \arg((w - z)/z)$. If the greatest possible $\tan^2 \theta$ is reduced p/q, find $p + q$.

复数满足 $|w| = 1, |z| = 10$，令 $\theta = \arg((w - z)/z)$。若 $\tan^2 \theta$ 的最大值为最简分数 p/q，求 $p + q$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 1.3 Complex numbers & roots of unity

复数与单位根

[AIME-2014-II-Q10] | Part B (Q6-Q10)

Fix complex z with $|z| = 2014$. A polygon has vertices z and every complex w satisfying $1/(z+w) = 1/z + 1/w$. Its area is $n\sqrt{3}$ for an integer n. Find $n \bmod 1000$.

固定模为二千零一十四的复数 z，以 z 及所有满足 $1/(z+w) = 1/z + 1/w$ 的复数 w 为顶点构成多边形。面积为整数 n 乘 $\sqrt{3}$，求 n 模 1000。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 1.3 Complex numbers & roots of unity

复数与单位根

[AIME-2016-I-Q07] | Part B (Q6-Q10)

For integers a, b, consider $z = [\sqrt{ab + 2016} - i\sqrt{|a + b|}]/(ab + 100)$, with the principal complex square root. Count ordered pairs for which the expression is defined and real.

对整数 a, b，考虑 $z = [\sqrt{ab + 2016} - i\sqrt{|a + b|}]/(ab + 100)$，根号取复数主平方根。求使表达式有定义且为实数的有序对数目。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 1.3 Complex numbers & roots of unity

复数与单位根

[AIME-2017-I-Q10] | Part B (Q6-Q10)

Let $z_1 = 18 + 83i, z_2 = 18 + 39i, z_3 = 78 + 99i$. Among complex $z \neq z_3$ for which $[(z_3 - z_1)/(z_2 - z_1)][(z - z_2)/(z - z_3)]$ is real, choose the one with largest imaginary part. Find its real part.

令 $z_1 = 18 + 83i, z_2 = 18 + 39i, z_3 = 78 + 99i$。在使 $[(z_3 - z_1)/(z_2 - z_1)][(z - z_2)/(z - z_3)]$ 为实数的 $z \neq z_3$ 中，选择虚部最大者。求其实部。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 1.3 Complex numbers & roots of unity

复数与单位根

[AIME-2018-I-Q06] | Part B (Q6-Q10)

Count unit complex numbers z for which $z^{720} - z^{120}$ is real. Find the count modulo 1000.

求单位复数 z 的个数，使 $z^{720} - z^{120}$ 为实数，取个数模 1000。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 1.3 Complex numbers & roots of unity

复数与单位根

[AIME-2019-II-Q08] | Part B (Q6-Q10)

Real coefficients $a, b, c \leq 2019$ define $f(z) = az^{2018} + bz^{2017} + cz^{2016}$ with $f((1+i\sqrt{3})/2) = 2015 + 2019i\sqrt{3}$. Find $f(1) \bmod 1000$.

实系数 $a, b, c \leq 2019$ 定义 $f(z) = az^{2018} + bz^{2017} + cz^{2016}$，且 $f((1+i\sqrt{3})/2) = 2015 + 2019i\sqrt{3}$。求 $f(1)$ 除以一千的余数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 1.3 Complex numbers & roots of unity

复数与单位根

[AIME-2020-I-Q08] | Part B (Q6-Q10)

Starting at O, a walker moves 5 units east. Each subsequent displacement is half the preceding length and is rotated 60° counterclockwise. The positions converge to P. If $OP^2 = m/n$ in lowest terms, find $m+n$.

从 O 出发，首段向东走 5 单位；以后每段长度为前段一半，方向相对前段逆时针转 60°。位置趋于 P。若 $OP^2 = m/n$ 已约分，求 $m+n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

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Section 1.3 Complex numbers & roots of unity

复数与单位根

[AIME-2023-II-Q08] | Part B (Q6-Q10)

Let $\omega = e^{2\pi i/7}$. Evaluate $\prod_{k=0}^{6}(1+\omega^k+\omega^{3k})$.

令 $\omega = e^{2\pi i/7}$，求 $\prod_{k=0}^{6}(1+\omega^k+\omega^{3k})$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 1.3 Complex numbers & roots of unity

复数与单位根

[AIME-2024-I-Q07] | Part B (Q6-Q10)

For complex z with $|z| = 4$, find the maximum real part of $(75 + 117i)z + (96 + 144i)/z$.

复数 z 满足 $|z| = 4$，求 $(75 + 117i)z + (96 + 144i)/z$ 实部的最大值。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 1.3 Complex numbers & roots of unity

复数与单位根

[AIME-2001-II-Q14] | Part C (Q11-Q15)

List the unit-modulus roots of $z^{28} - z^8 - 1 = 0$ by increasing arguments $0 \le \theta_1 < \cdots < \theta_{2n} < 360^\circ$. Find $\theta_2 + \theta_4 + \cdots + \theta_{2n}$.

将 $z^{28} - z^8 - 1 = 0$ 的单位模根按辐角 $0 \le \theta_1 < \cdots < \theta_{2n} < 360^\circ$ 排序，求偶下标辐角之和。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 1.3 Complex numbers & roots of unity

复数与单位根

[AIME-2002-I-Q12] | Part C (Q11-Q15)

Let $F(z) = (z+i)/(z-i)$ and $z_n = F(z_{n-1})$, starting from $z_0 = 1/137 + i$. If $z_{2002} = a + bi$, find $a + b$.

令 $F(z) = (z+i)/(z-i)$，$z_n = F(z_{n-1})$，初值 $z_0 = 1/137 + i$。若 $z_{2002} = a + bi$，求 $a + b$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

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Section 1.3 Complex numbers & roots of unity

复数与单位根

[AIME-2003-II-Q15] | Part C (Q11-Q15)

Let $P(x) = 24x^{24} + \sum_{j=1}^{23}(24-j)(x^{24-j} + x^{24+j})$. For each distinct zero z_k, write $z_k^2 = a_k + ib_k$. If $\sum_k |b_k| = m + n\sqrt{p}$ with squarefree p, find $m+n+p$.

令 $P(x) = 24x^{24} + \sum_{j=1}^{23}(24-j)(x^{24-j} + x^{24+j})$。对每个不同零点写 $z_k^2 = a_k + ib_k$。若 $\sum_k |b_k| = m + n\sqrt{p}$，$p$ 无平方因子，求参数和。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 1.3 Complex numbers & roots of unity

复数与单位根

[AIME-2004-I-Q13] | Part C (Q11-Q15)

The 34 roots of $P(x) = (1 + x + \cdots + x^{17})^2 - x^{17}$ have arguments $2\pi\alpha_k$ ordered by $0 < \alpha_1 \le \cdots \le \alpha_{34} < 1$. If $\alpha_1 + \cdots + \alpha_5 = m/n$ reduced, find $m + n$.

多项式 $P(x) = (1 + x + \cdots + x^{17})^2 - x^{17}$ 的三十四根辐角为 $2\pi\alpha_k$，按 $0 < \alpha_1 \le \cdots \le \alpha_{34} < 1$ 排序。若前五个 α 和约分为 m/n，求 $m + n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 1.3 Complex numbers & roots of unity

复数与单位根

[AIME-2008-II-Q13] | Part C (Q11-Q15)

A regular hexagon centered at zero in the complex plane has opposite sides distance one apart, with one pair vertical. Let R be its exterior and $S = \{1/z : z \in R\}$. If S has area $a\pi + \sqrt{b}$ for positive integers, find $a + b$.

复平面正六边形中心为零，对边距离为一，其中一对竖直。令 R 为其外部，$S = \{1/z : z \in R\}$。若面积为 $a\pi + \sqrt{b}$，求 $a + b$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 1.3 Complex numbers & roots of unity

复数与单位根

[AIME-2009-II-Q13] | Part C (Q11-Q15)

Divide a radius-two semicircle from endpoint A to endpoint B into seven equal arcs by points $C_1, \ldots, C_6$. Multiply all twelve lengths AC_i, BC_i to obtain N. Find $N \bmod 1000$.

半径二的半圆弧端点为 A, B，用六点 C_i 分成七等弧。十二条弦 AC_i, BC_i 长度之积为 N，求其除以一千的余数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 1.3 Complex numbers & roots of unity

复数与单位根

[AIME-2012-I-Q14] | Part C (Q11-Q15)

The roots a, b, c of $z^3 + qz + r$ are vertices of a right triangle in the complex plane. Given $|a|^2 + |b|^2 + |c|^2 = 250$, find the squared hypotenuse length.

多项式 $z^3 + qz + r$ 的三个复根在复平面构成直角三角形，且模平方和为 250。求斜边长的平方。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

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Section 1.3 Complex numbers & roots of unity

复数与单位根

[AIME-2013-I-Q14] | Part C (Q11-Q15)

For $\pi \le \theta < 2\pi$, define $P = \sum_{k\ge 1} 2^{-k} \sin(k(\theta + \pi/2))$ and $Q = 1 + \sum_{k\ge 1} 2^{-k} \cos(k(\theta + \pi/2))$. Given $P/Q = 2\sqrt{2}/7$, write $\sin\theta = -m/n$ in lowest terms and find $m + n$.

$\pi \le \theta < 2\pi$，定义 $P = \sum_{k\ge 1} 2^{-k} \sin(k(\theta + \pi/2))$，$Q = 1 + \sum_{k\ge 1} 2^{-k} \cos(k(\theta + \pi/2))$。若 $P/Q = 2\sqrt{2}/7$，将 $\sin\theta = -m/n$ 约为最简，求 $m + n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

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Section 1.3 Complex numbers & roots of unity

复数与单位根

[AIME-2017-II-Q12] | Part C (Q11-Q15)

Circle C_0 has radius1 and marked point A_0. For $0 < r < 1$, circle C_n has radius r^n and is internally tangent to C_{n-1} at A_{n-1}. Mark A_n by rotating the radius from its center to A_{n-1} counterclockwise by90 degrees. All these disks have one common point B. When $r = 11/60$, express the distance from the center of C_0 to B as reduced m/n and find $m + n$.

圆 C_0 半径为1，标记点为 A_0。对 $0 < r < 1$，圆 C_n 半径为 r^n，并在 A_{n-1} 与 C_{n-1} 内切。将从其圆心指向 A_{n-1} 的半径逆时针旋转90度得到 A_n。所有闭圆盘共有一点 B。当 $r = 11/60$，将 C_0 圆心到 B 的距离写成最简分数 m/n，求 $m + n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 1.3 Complex numbers & roots of unity

复数与单位根

[AIME-2019-I-Q12] | Part C (Q11-Q15)

Let $f(z) = z^2 - 19z$. A value $z = m + \sqrt{n} + 11i$ with positive integers m, n makes $z, f(z), f(f(z))$ a nondegenerate right triangle with right angle at $f(z)$. Find $m + n$.

令 $f(z) = z^2 - 19z$。某 $z = m + \sqrt{n} + 11i$（m, n 为正整数）使 $z, f(z), f(f(z))$ 成非退化直角三角形，直角在 $f(z)$。求 $m + n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 1.3 Complex numbers & roots of unity

复数与单位根

[AIME-2023-I-Q15] | Part C (Q11-Q15)

Find the largest prime $p < 1000$ for which a Gaussian integer z has $|z| = \sqrt{p}$ and the three positive numbers $p, \mathrm{Re}(z^3), \mathrm{Im}(z^3)$ are triangle side lengths.

求小于 1000 的最大素数 p，使存在实部、虚部均为整数的复数 z，满足 $|z| = \sqrt{p}$，且 $p, \mathrm{Re}(z^3), \mathrm{Im}(z^3)$ 均为正并可作三角形三边。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

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Section 1.3 Complex numbers & roots of unity

复数与单位根

[AIME-2024-II-Q13] | Part C (Q11-Q15)

For a nontrivial thirteenth root of unity ω, find $\prod_{k=0}^{12}(2-2\omega^k+\omega^{2k})$ modulo 1000.

设 $\omega \neq 1$ 为十三次单位根，求 $\prod_{k=0}^{12}(2-2\omega^k+\omega^{2k})$ 除以 1000 的余数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 1.3 Complex numbers & roots of unity

复数与单位根

[AIME-2026-II-Q11] | Part C (Q11-Q15)

Find the greatest integer n for which $x^3 - (n/6)x^2 + (n-11)x - 400$ has roots $\alpha^2, \beta^2, \gamma^2$ allowing exactly seven distinct sums $\alpha + \beta + \gamma$.

求最大整数 n，使三次多项式 $x^3 - (n/6)x^2 + (n-11)x - 400$ 的根为 $\alpha^2, \beta^2, \gamma^2$ 时，$\alpha + \beta + \gamma$ 恰有七个不同可能值。

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